

MAREEBA, Australia

WMO: 952860

Lat: 17.067S

Lon: 145.428E

Elev: 1552

StdP: 13.89

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.7	54.2	38.2	35.9	67.9	42.1	41.8	64.9	20.8	66.9	19.0	67.9	6.5	140	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	17.2	91.7	70.8	89.4	71.0	87.6	70.9	76.3	83.2	75.5	82.0	74.8	81.1	9.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	137.8	78.0	74.0	134.3	77.6	73.3	131.3	77.2	41.1	83.6	40.3	82.1	39.6	81.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.4	15.8	DB	47.3	97.4	2.7	2.4	45.4	99.1	43.8	100.5	42.3	101.8	40.4	103.5	(4)
(5)			WB	43.8	78.3	2.3	1.7	42.1	79.5	40.7	80.5	39.4	81.4	37.7	82.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.2	78.5	78.0	76.5	73.9	70.7	67.9	66.4	67.3	70.8	74.1	76.6	78.6	(6)
(7)		DBStd	5.10	2.41	2.19	2.18	2.38	2.70	3.08	3.12	3.08	2.87	2.91	2.47	2.45	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	48	0	0	0	0	1	11	22	13	1	0	0	0	(9)
(10)		CDD50	8483	882	784	821	718	641	537	509	535	623	748	799	886	(10)
(11)		CDD65	3057	417	364	356	268	178	98	65	84	174	283	349	421	(11)
(12)		CDH74	19076	2997	2458	2082	1265	579	250	172	375	1091	1949	2560	3298	(12)
(13)		CDH80	5405	969	720	488	198	48	14	8	33	234	596	869	1228	(13)
(14)	Wind	WSAvg	7.6	6.1	5.9	7.4	8.8	8.7	8.9	8.2	7.3	7.4	7.0	6.4	(14)	
(15)	Precipitation	PrecAvg	37.6	8.1	9.4	7.4	1.8	1.1	0.9	0.6	0.5	0.4	0.9	2.2	4.4	(15)
(16)		PrecMax	63.1	20.6	25.3	17.7	10.0	3.3	3.2	1.6	3.5	1.3	4.2	8.0	14.3	(16)
(17)		PrecMin	21.1	2.3	1.2	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	(17)
(18)		PrecStd	11.2	4.3	6.1	4.8	2.2	0.9	0.7	0.5	0.7	0.4	0.9	2.3	3.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.9	92.2	89.3	86.4	83.8	81.8	80.8	83.4	88.6	92.6	94.0	94.5	(19)
(20)			MCWB	71.6	72.6	72.1	69.9	68.2	67.2	64.9	65.0	64.2	66.5	69.1	70.6	(20)
(21)		2%	DB	89.9	88.9	86.6	84.1	81.2	78.9	77.9	80.5	85.1	88.5	89.6	91.1	(21)
(22)			MCWB	72.7	73.4	72.5	70.1	67.6	66.3	64.3	64.1	64.7	67.3	68.9	71.5	(22)
(23)		5%	DB	87.5	86.6	84.7	82.3	79.2	76.8	75.8	78.1	82.7	85.7	87.1	88.7	(23)
(24)			MCWB	72.6	73.2	72.0	69.6	67.2	65.4	63.5	63.5	64.9	67.4	69.4	71.4	(24)
(25)		10%	DB	85.3	84.5	82.8	80.6	77.3	74.8	73.7	75.9	80.2	83.3	85.0	86.4	(25)
(26)			MCWB	72.5	73.2	71.8	69.2	66.6	64.5	62.7	62.7	64.6	67.3	69.2	71.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	77.2	77.0	75.0	71.8	70.3	69.1	69.9	71.6	73.4	75.4	76.9	(27)
(28)			MCDWB	84.7	84.8	83.8	82.0	78.9	77.0	75.0	78.3	81.4	83.0	84.3	83.7	(28)
(29)		2%	WB	76.1	76.3	75.4	73.5	70.3	68.6	67.2	67.5	69.7	71.8	73.6	75.4	(29)
(30)			MCDWB	82.8	83.0	81.4	79.7	76.9	75.0	73.6	75.5	78.5	80.9	81.9	82.8	(30)
(31)		5%	WB	75.2	75.5	74.5	72.3	69.3	67.2	65.8	66.0	68.2	70.5	72.5	74.5	(31)
(32)			MCDWB	81.6	81.4	80.2	78.2	75.6	73.6	72.3	74.2	76.9	79.5	80.9	81.9	(32)
(33)		10%	WB	74.5	74.7	73.7	71.1	68.3	66.1	64.6	64.6	66.7	69.2	71.5	73.7	(33)
(34)			MCDWB	80.4	80.3	79.2	77.0	74.6	72.6	71.3	72.5	75.6	78.3	79.9	80.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	13.9	13.3	14.1	14.6	15.4	16.3	18.4	20.7	20.0	18.8	17.2	(35)
(36)			MCDBR	19.1	17.8	16.9	17.0	17.1	17.9	18.8	21.8	24.5	24.0	22.2	21.2	(36)
(37)			MCWBR	5.4	5.4	5.6	5.8	6.2	7.2	7.4	7.9	8.1	7.0	6.1	5.7	(37)
(38)		5% WB	MCDBR	14.4	13.8	13.2	13.0	13.6	14.5	14.1	18.1	19.7	19.2	18.3	16.4	(38)
(39)			MCWBR	5.2	5.1	5.4	5.4	5.9	6.6	6.4	7.4	7.7	6.7	5.9	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.377	0.375	0.363	0.352	0.338	0.330	0.315	0.326	0.353	0.371	0.377	0.385	(40)	
(41)		taud	2.577	2.591	2.612	2.617	2.618	2.621	2.655	2.601	2.501	2.476	2.499	2.512	(41)	
(42)		Ebn at Noon	307	304	301	293	286	283	292	297	299	302	305	304	(42)	
(43)		Edh at Noon	34	33	31	29	27	26	26	29	35	37	36	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1838	1766	1651	1515	1340	1211	1333	1611	1936	2090	2128	2036	(44)	
(45)		RadStd	144	206	146	146	151	117	116	98	133	113	152	174	231	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page